MME 351 - Marine Electrical Technology

(2 0 2)

E.A. Frimpong (PhD), MIEEE, MGhIE

Associate Professor at Department of Electrical and Electronic Engineering,
Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Course aim

- To provide insight into electrical technology used in marine vessels.







Course Content

- **Power generation and distribution:** Selection of DC and AC generators, Merits and Demerits of AC and DC. Construction & Characteristics of brushless high speed AC generators with automatic Voltage regulator.
- **Rules and Regulations governing electrical machinery on ships**, location and installation of generator sets. Rating and characteristics of main switch boards. Arrangements of preferential trips and protective devices with wiring diagrams.
- **System earthing:** Isolated neutral systems, resistance earthed system.
- **Motor and Control Equipment:** Characteristics of continuous maximum rated machines. Centralized control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.

Course Content

- **Motor and Control Equipment:** Special requirements of motors and starters for anchor Windlass & Capstan, wiring diagram, lowering & hoisting arrangement, overload protection, magnetic disc, brakes. Centralised control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.
- **Starters for miscellaneous electrical equipment** for machinery space auxiliaries eg. Sequential starting and cut outs for an automatic fired boiler incorporating safety devices & combustion control equipment. Sequential starting for refrigerating plants, incorporating safety devices like high pressure cut out, cooling water failure cut out, automatic defrosting with help of timer.

Assessment

- **Continuous assessment** (Assignments, quizzes, mid-sem) - 30%
- **End-of-Semester Examination** (MCQs, Written paper) - 70%

Power Generation and Distribution in Marine Vessels



Uses of electricity in watercraft



Lighting



Pumps

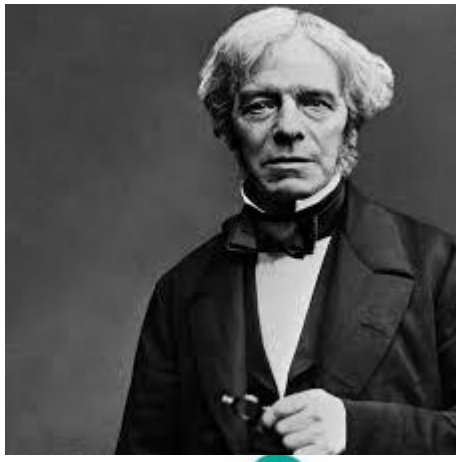


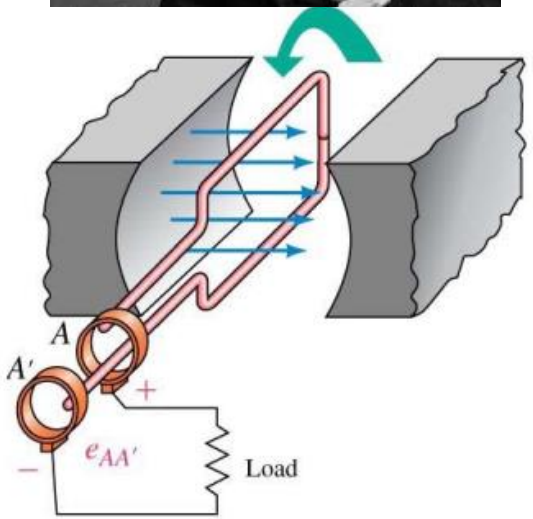


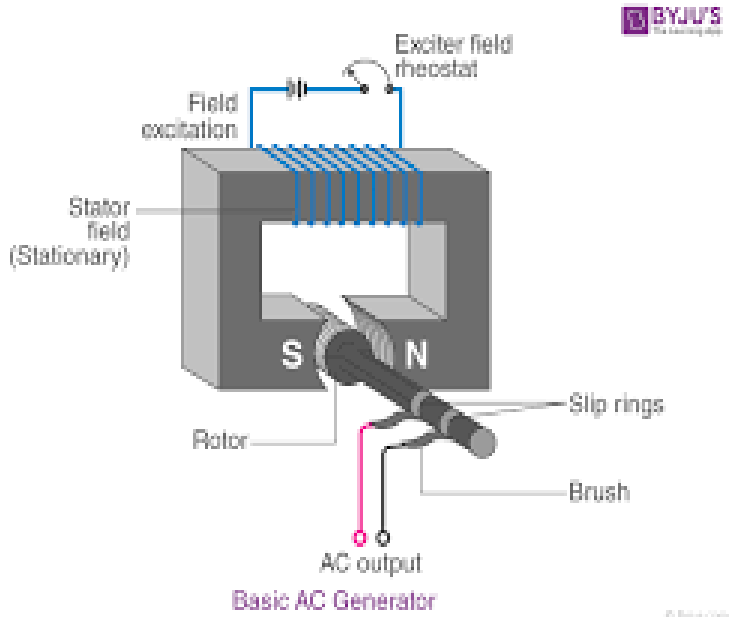
Refrigeration

AC generators

Operating principle of AC generators





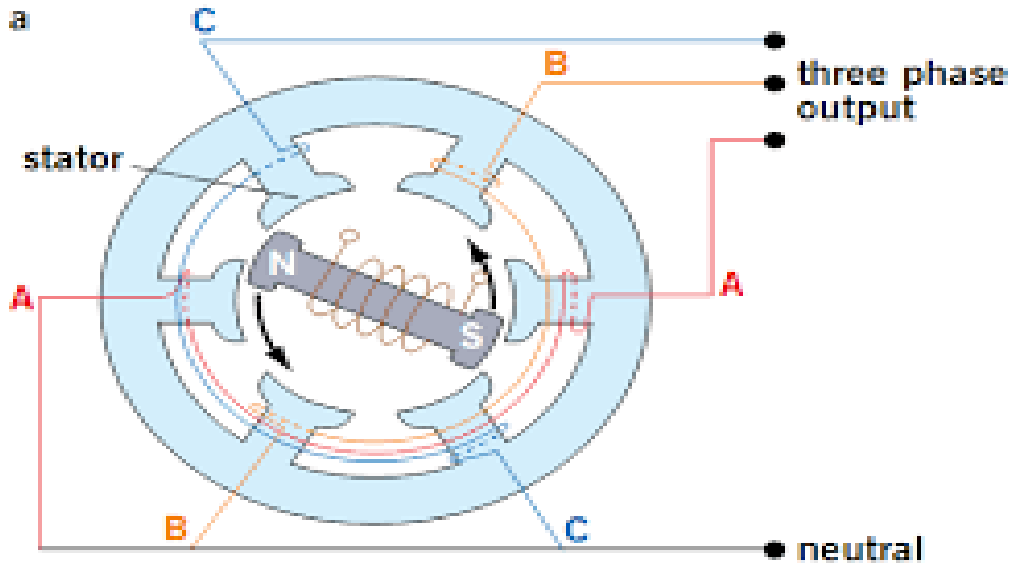




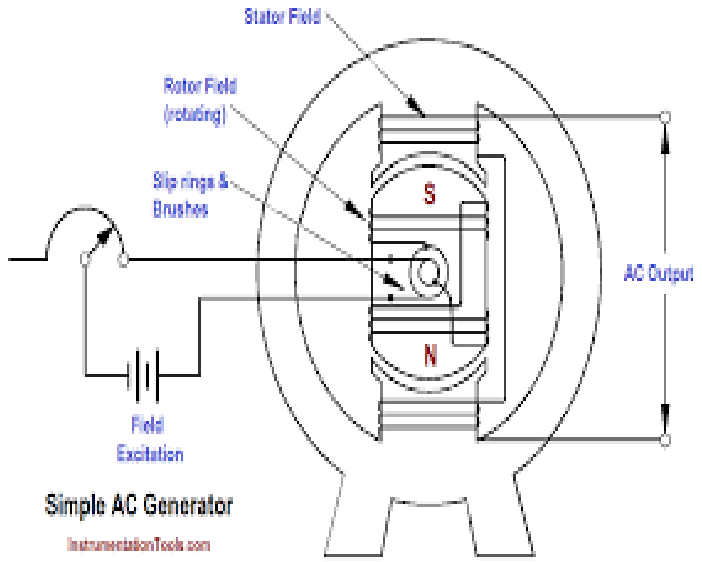


AC generators

Types AC generators



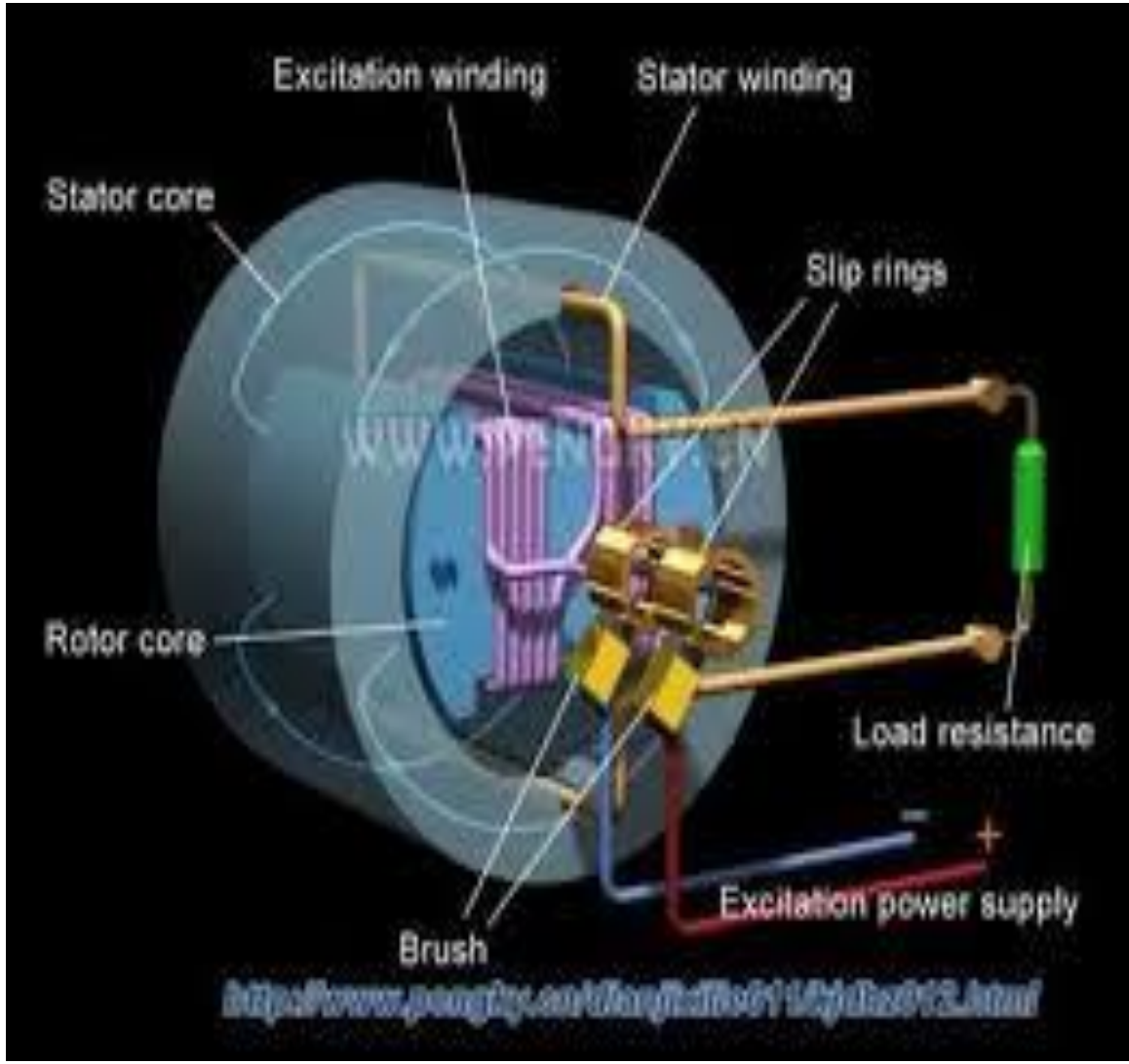
3-phase AC generator



1-phase AC generator

AC generators

- **Major components**
  - **Stator**
  - **Rotor**
  - **Slip rings**

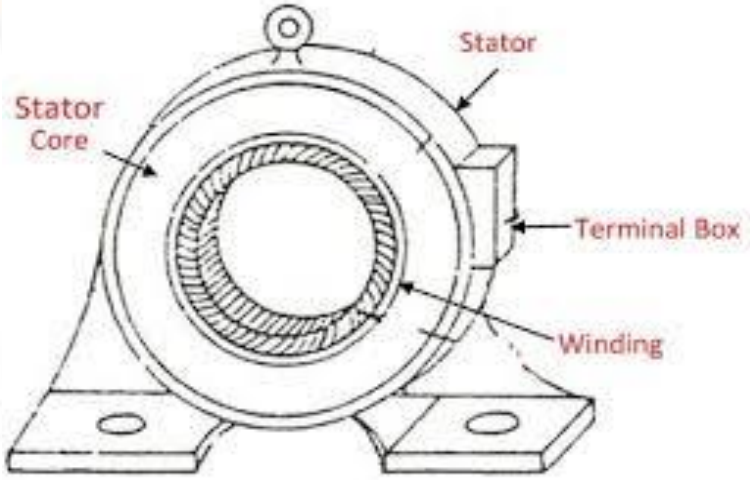


AC generators

- **Stator**











AC generators

- **Rotor**

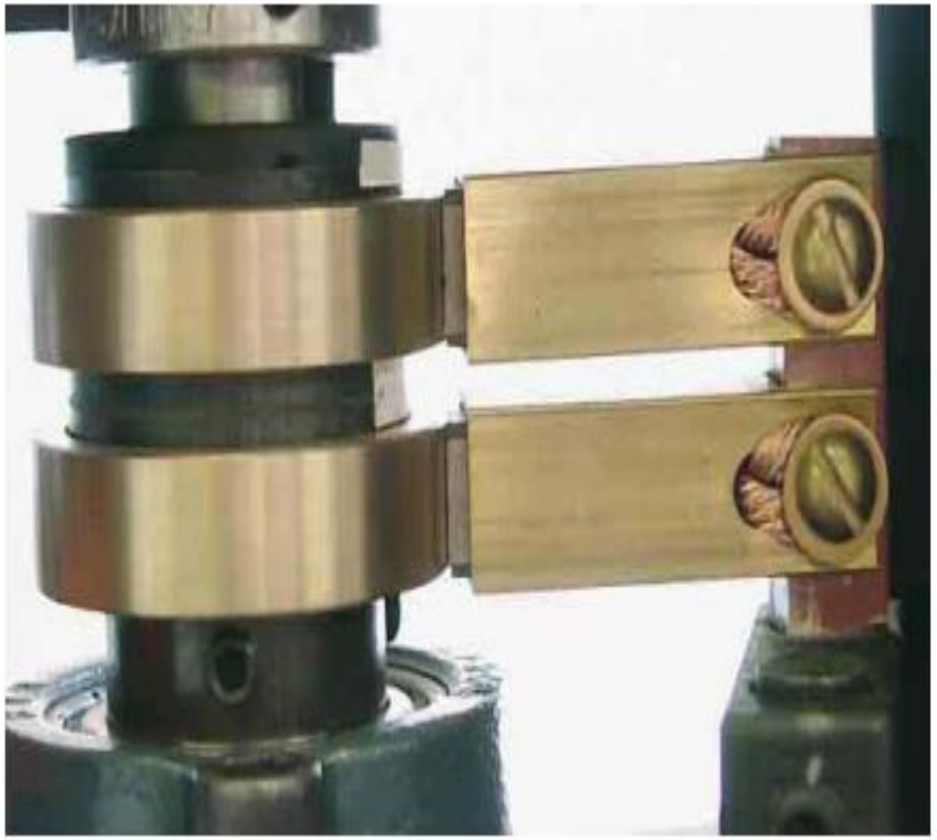






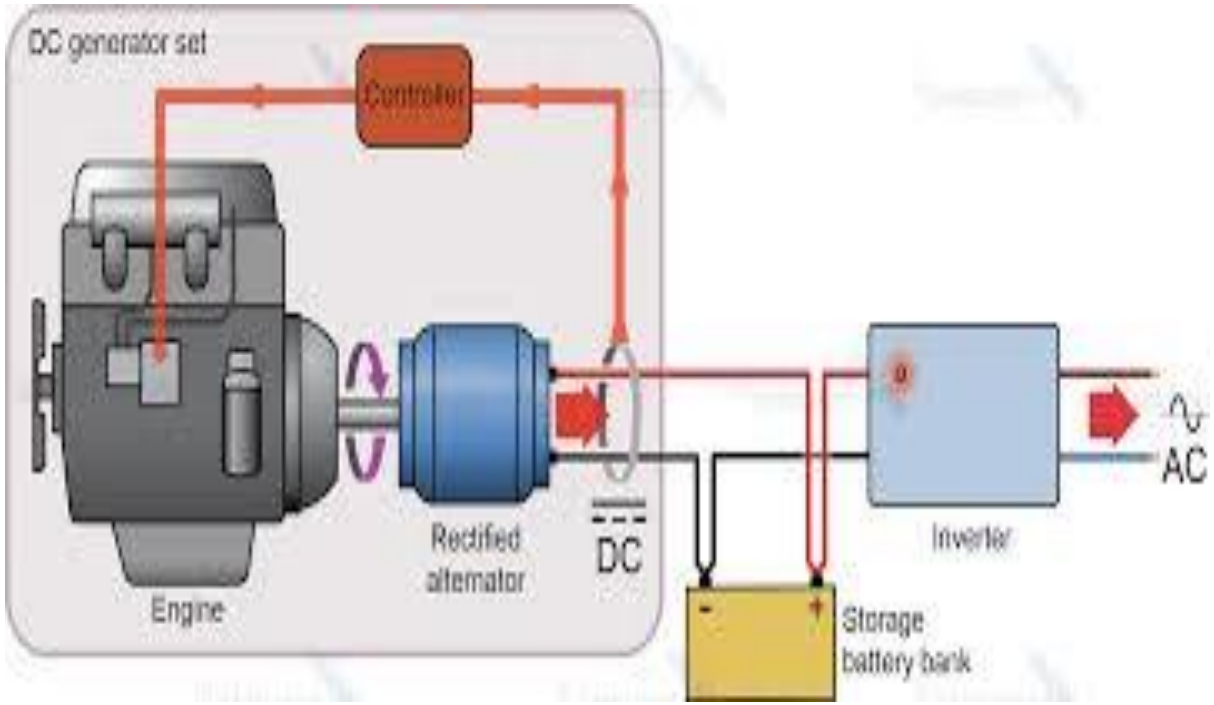
AC generators

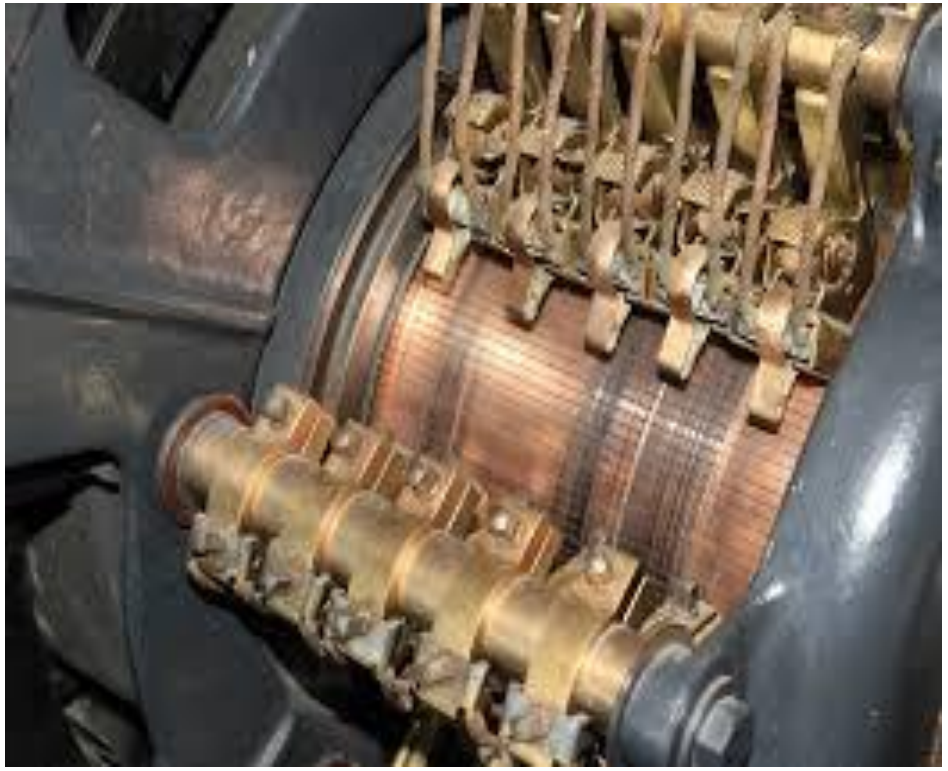
- **Slip rings**



DC generators

- **A DC generator is essentially a 1-phase AC generator with commutator**





AC vrs DC generators

- **DC Generators**
  - Suitability for large motors
  - Simple design
- **AC Generators**
  - Suitability for electrical appliances and small motors
  - Low maintenance needs
  - Easy distribution of the output with a transformer
  - Higher efficiency
  - Losses are relatively fewer than DC Generators
  - AC generators can be easily stepped up and stepped down through transformers.
  - Smaller in sizer compared to DC generators.

Emergency power supply

- Emergency power is needed when the main supply system fails.
- Emergency power can be supplied by
  - Batteries
  - Emergency generator
  - Both batteries and generator
- Emergency power should cover the following key components and systems
  - Steering gear system
  - Emergency bilge pump
  - Watertight doors
  - Fire fighting system
  - Ship's navigation lights and emergency lights
  - Communication and alarm system



Location and installation of generators

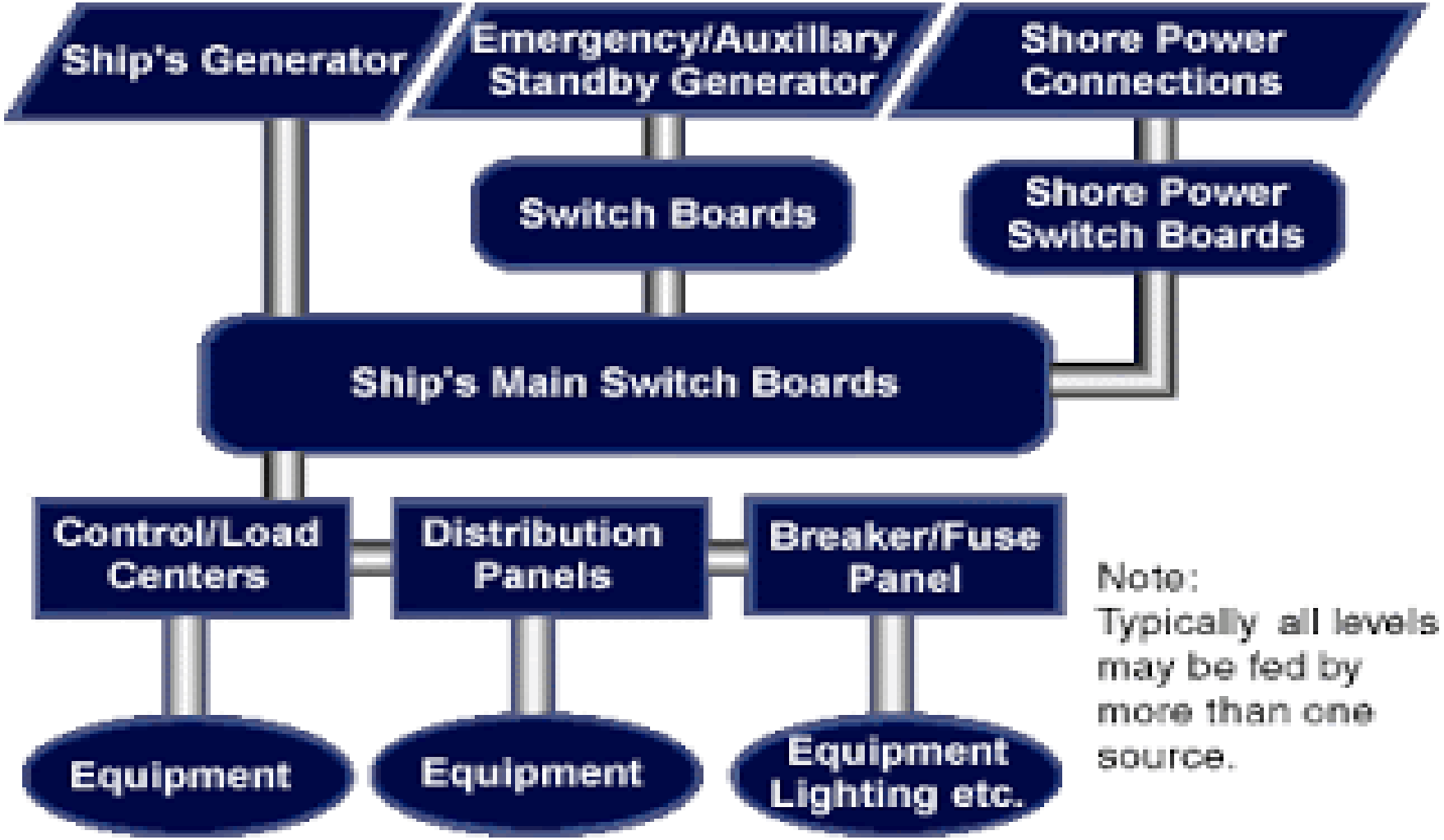
- A solid, level mounting platform is necessary
- Adequate space on all sides for ventilation and servicing
- Generators must be kept away from living quarters, and away from bilge splash and vapors.

Emergency power supply

- **Emergency generator is normally located outside the machinery space to provided unhindered access to it.**

Electrical distribution systems

- The power distribution system of a ship is designed to provide a secure supply to all loads with built-in protection for equipment and personnel.



Electrical distribution systems

A ship's electricity distribution system consists of different component for distribution and safe operation of the system. They are:

- Ship generator consisting of prime mover and alternator.
- Main switch board which is a metal enclosure taking power from a generator and supplying it to different machinery.
- Bus Bars which act as a carrier and allow transfer for transfer of power from one point to another.
- Circuit breakers which act as a switch and in unsafe condition can be tripped to avoid breakdown and accidents.
- Fuses as safety device for machinery.
- Transformers to step up or step down the voltage.

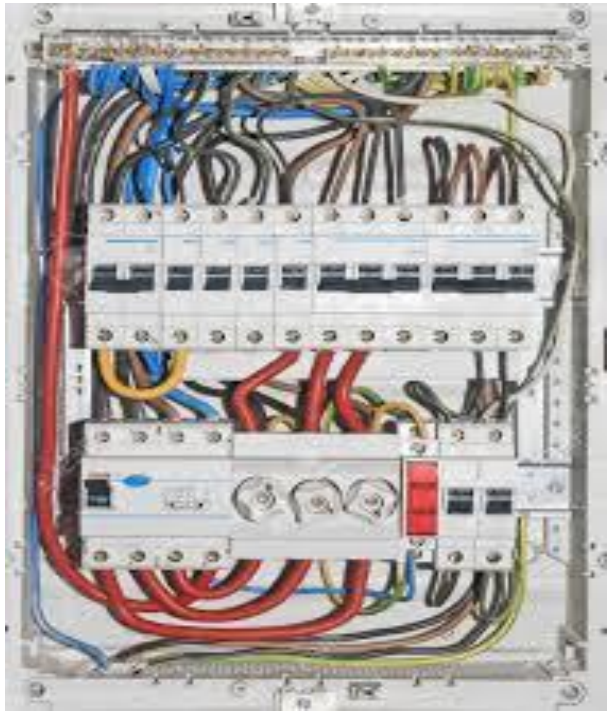
Electrical distribution systems

- **Generator**



Electrical distribution systems

- **Main switch board**



Domestic



Ship

Electrical distribution systems

- **Busbars**







Electrical distribution systems

- **Fuse**



Electrical distribution systems

- **Circuit breaker**







Electrical distribution systems

- **Transformer**



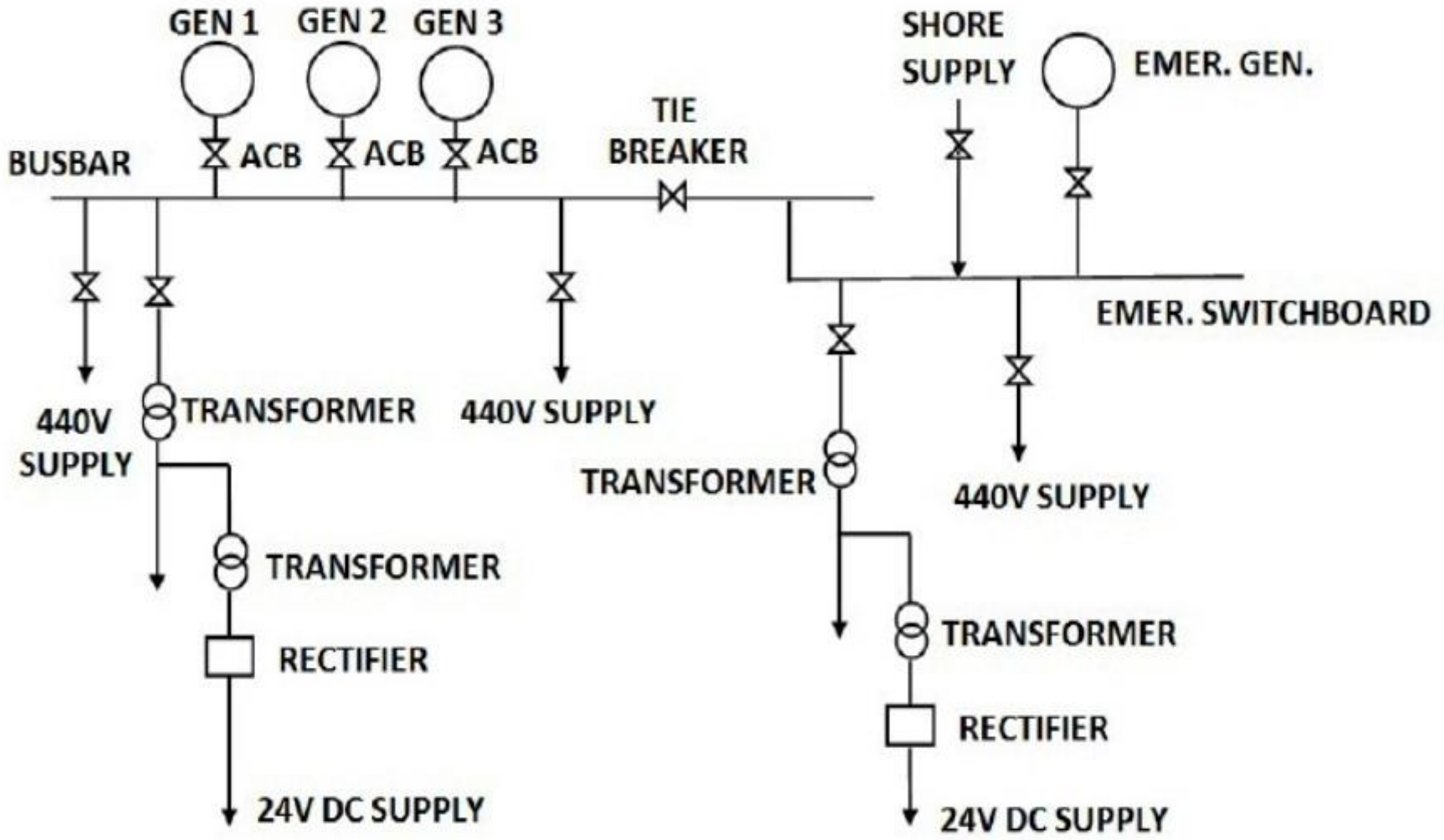








Electrical distribution systems



A typical distribution system for a ship

Emergency switchboard

- It is the emergency power distribution center.
- It distributes power to auxiliaries that are vital, under emergency conditions
- Under normal operation, power for the emergency switchboard is supplied by the main generators through a bus-tie.
- In case of power failure, loss of voltage at the emergency switchboard will cause the emergency generator to automatically start.
- Simultaneously, the bus-tie will open disconnecting the main and the emergency switchboards.
- The emergency generator breaker will close, energizing the emergency board and providing the needed emergency power.

Electrical distribution systems

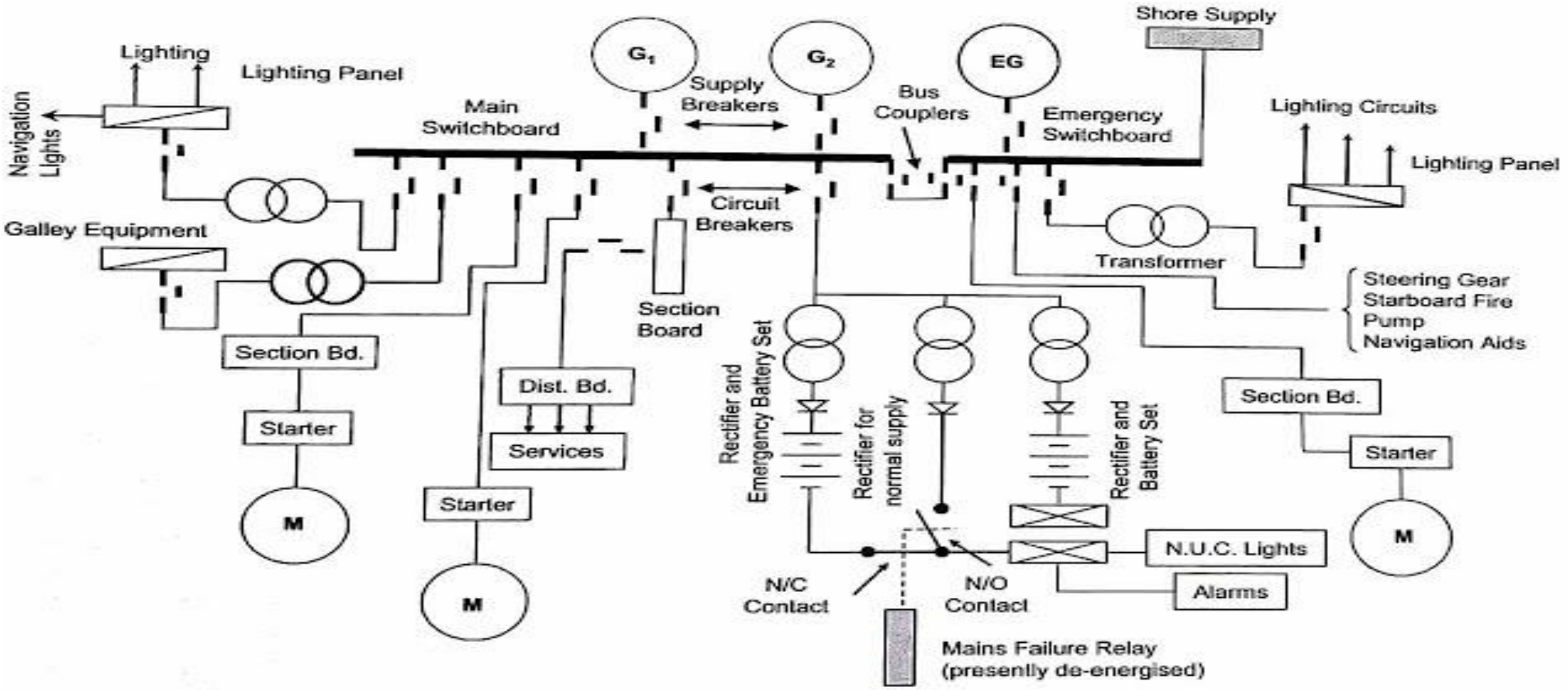
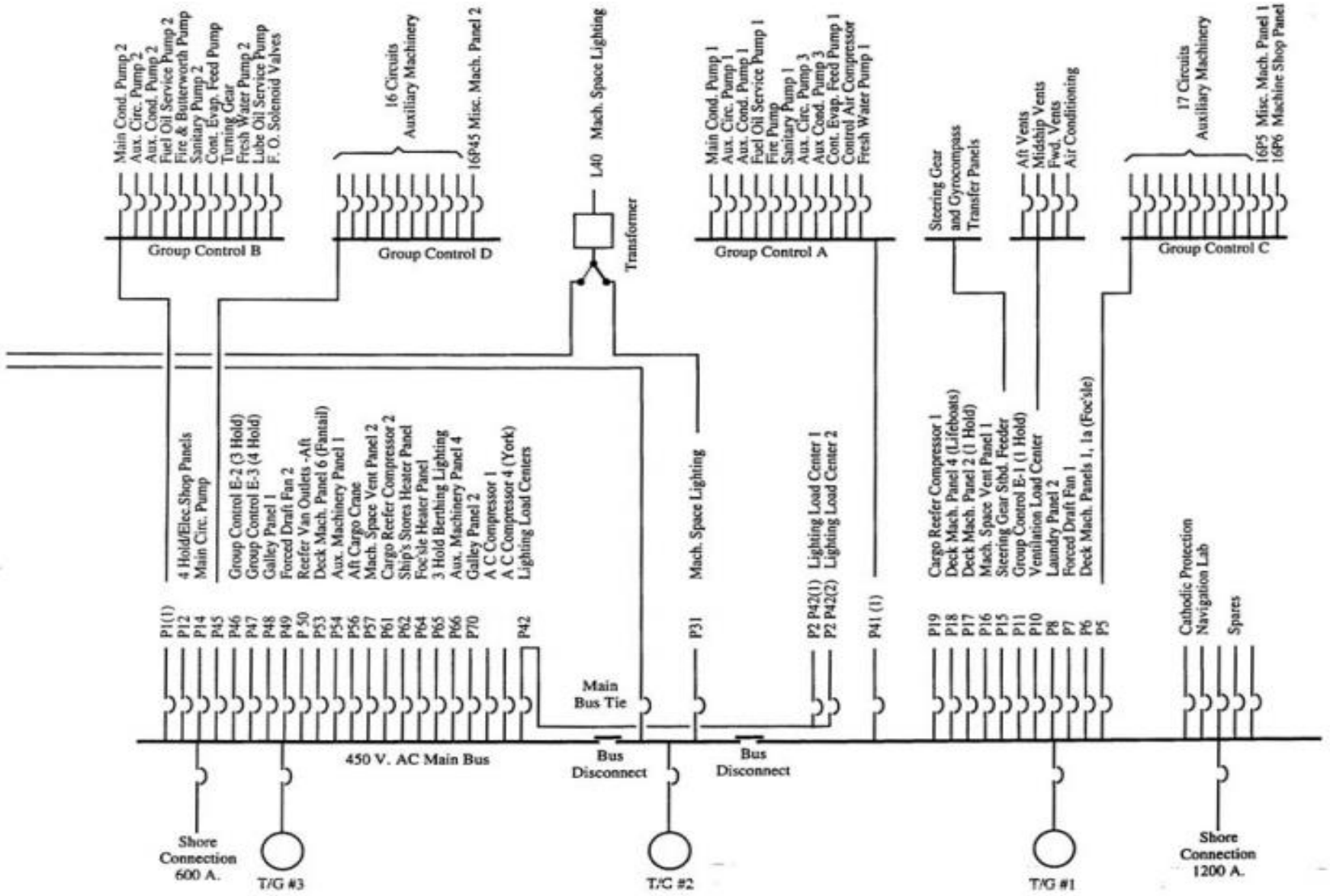
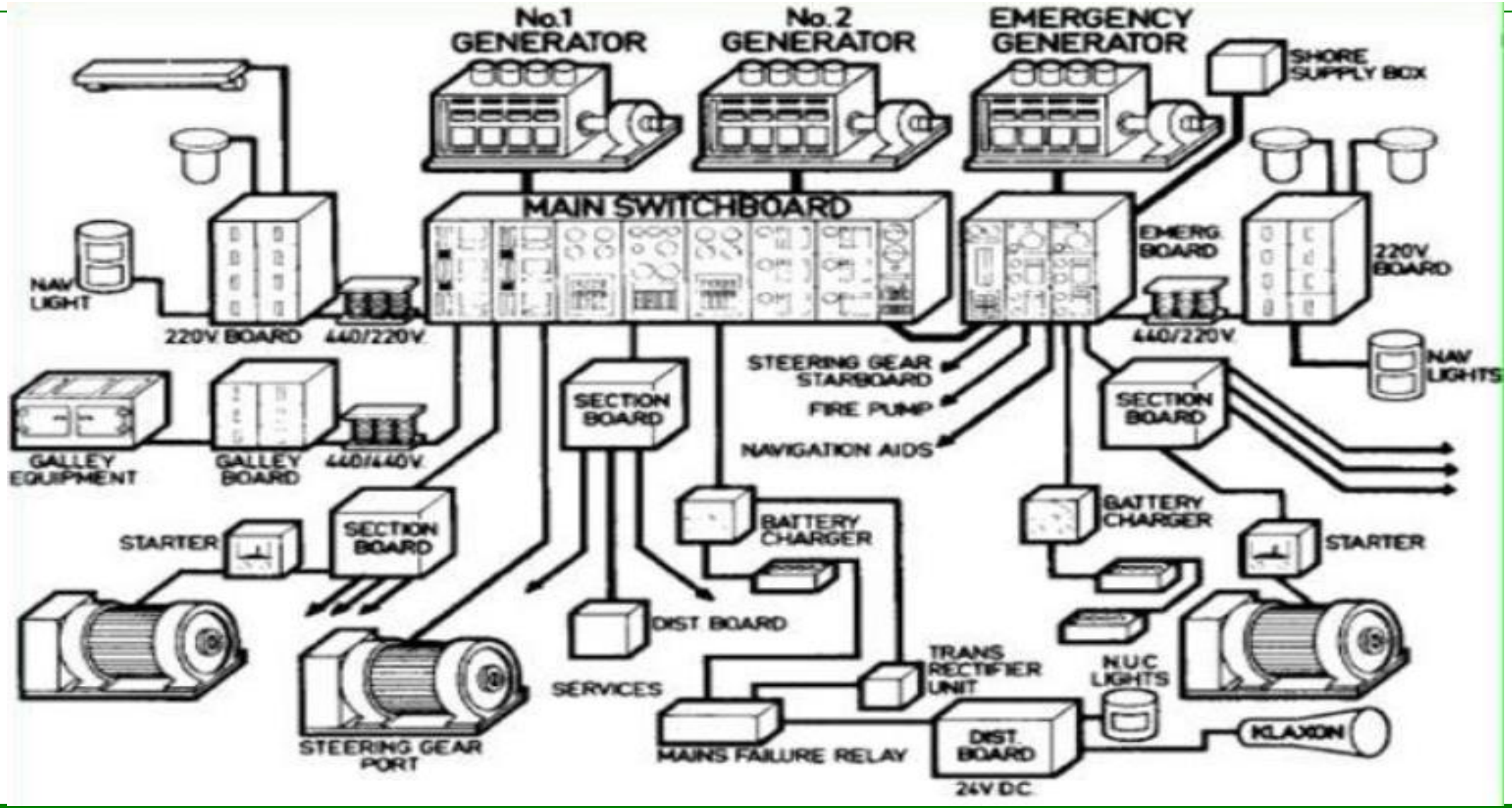


Figure 4.9 – A Ship's Typical Electrical Distribution System



Ship's distribution system


No.1 GENERATOR
No.2 GENERATOR
EMERGENCY GENERATOR
SHORE SUPPLY BOX
MAIN SWITCHBOARD
NAV LIGHT
220V BOARD
440/220V
EMERG BOARD
220V BOARD
NAV LIGHTS
STEERING GEAR STARBOARD
FIRE PUMP
NAVIGATION AIDS
GALLEY EQUIPMENT
GALLEY BOARD
440/440V
SECTION BOARD
SECTION BOARD
BATTERY CHARGER
BATTERY CHARGER
STARTER
SECTION BOARD
STARTER
DIST BOARD
SERVICES
TRANS RECTIFIER UNIT
NUC LIGHTS
KLAXON
STEERING GEAR PORT
MAINS FAILURE RELAY
DIST BOARD
24V DC

Electrical distribution systems



Ship shore supply